

Minh-Tâm Quang Trinh

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Citizenship

American

Current Research

[algebraic geometry](#), [combinatorics](#), [quantum algebra](#), [representation theory](#)

Employment

Assistant Professor (tenure-track). [Howard University](#) 2025–
Gibbs Assistant Professor. [Yale University](#) 2024–25
CLE Moore Instructor. [Massachusetts Institute of Technology](#) 2020–24
NSF Mathematical Sciences Postdoctoral Research Fellow (2020–2023)
— *Sponsoring Scientist*: Zhiwei Yun

Education

PhD in Mathematics. [University of Chicago](#) 2014–20
— *Primary Advisor*: Châu Ngô (= Ngô Bảo Châu)
— *Secondary Advisor*: Victor Ginzburg
William Rainey Harper Dissertation Fellow
AB in Mathematics magna cum laude. [Princeton University](#) 2010–14

Publications and Preprints

- Free Immersions in Critical Dimension from Lie Groups
Preprint. <https://mqtrinh.github.io/math/research/free.pdf>
- Haiman's Conjecture and Springer's Representations
Preprint. [arXiv:2605.20131](#)
- Partial Resolutions and Noncrossing Combinatorics
Preprint. [arXiv:2601.17293](#)
+ N. Williams
- Level-Rank Dualities from Φ -Cuspidal Pairs and Affine Springer Fibers
Preprint. [arXiv:2311.17106](#)
+ T. Xue
- Unipotent Elements and Kálmán–Serre Duality
Preprint. [arXiv:2210.09051](#)

- From the Hecke Category to the Unipotent Locus
Preprint. [arXiv:2106.07444](https://arxiv.org/abs/2106.07444)
- The Hilb-vs-Quot Conjecture
J. angew. reine Math. (Crelle), **828** (2025), 83–126.
+ O. Kivinen
- Simple Braids Tend Toward Positive Entropy
J. Knot Theory Ramifications, **33**(11) (2024), 2450034, 12 pp.
+ L. Robitaille
- Rational Noncrossing Coxeter–Catalan Combinatorics
Proc. Lond. Math. Soc. **129**(4) (2024), e12643, 50 pp.
+ P. Galashin, T. Lam, N. Williams
- Algebraic Braids and Geometric Representation Theory
PhD thesis. University of Chicago. ProQuest LLC (2020), 239 pp.
- Extremal Primes for Elliptic Curves
J. Num. Theory, **164** (2016), 282–298.
+ K. James, B. Tran, P. Wertheimer, D. Zantout
- Zeros of Dirichlet L -Functions over Function Fields
Comm. Num. Theory Phys., **8**(3) (2014), 511–539.
+ J. Andrade, S. J. Miller, K. Pratt

Extended Abstracts

- Level-Rank Dualities for Finite Reductive Groups
To appear in *Math. Res. Lett.* (2026). [arXiv:2508.07051](https://arxiv.org/abs/2508.07051)
+ T. Xue
- Cell Decompositions of Hecke Traces and Link Polynomials
Proceedings of the 37th International Conference on Formal Power Series and Algebraic Combinatorics (FPSAC), Sapporo, 2025. *Sém. Lothar. Combin.*, **93B** (2025), Paper No. 90.
+ N. Williams

Non-Research Publications

- The Hodge Theory of Soergel Bimodules
Chapter 18 in *Introduction to Soergel Bimodules*. Ed. B. Elias, S. Makisumi, U. Thiel, G. Williamson. RSME Springer Series, **5** (2020), 347–367.
+ L. Taylor

Invited Talks

- [KIAS–PKU–SKKU Combinatorics and Representation Theory Seminar](#). 12/11/26.
- [UT-Austin Geometry Seminar](#). 12/3/26.
- [SIAM TXLA Sectional Meeting](#). Minisymposium on Algebraic and Geometric Structures in Discrete Mathematics. 20-minute talk. SMU. TBA.

- [Workshop II: Contact Geometry, Cluster Algebras, and Skein Theory](#). Quantum Topology, Character Varieties and Low-Dimensional Geometry. Institute for Pure & Applied Mathematics (IPAM). TBA.
- [AMS Fall Eastern Sectional Meeting. Special Session on Knots and Braids](#). 20-minute talk. GWU. TBA.
- [AMS Fall Eastern Sectional Meeting. Special Session on Categorification, Low-Dimensional Topology, and Representation Theory](#). 50-minute talk. GWU. TBA.
- [MIT Lie Groups Seminar](#). 4/29/26.
- [Knots in Washington 53](#). 4/25/26. Plenary talk.
- [UT-Dallas Algebra and Combinatorics Seminar](#). 3/19/26.
- [GWU–Howard Combinatorics & Algebra Seminar](#). 2/10/26.
- [Knots in Washington 52](#). 12/7/25. 20-minute talk.
- [UMD Algebraic Geometry Seminar](#). 12/1/25.
- [UT-Dallas Algebra and Combinatorics Seminar](#). 11/19/25.
- [Howard University Mathematics Colloquium](#). 9/19/25.
- [USNA Mathematics Colloquium](#). 4/17/25.
- [Johns Hopkins Topology Seminar](#). 4/16/25.
- [Rice Algebraic Geometry and Number Theory Seminar](#). 4/1/25.
- [UChicago Representation Theory Seminar](#). 1/7/25.
- [Northeastern Geometry, Physics, and Representation Theory Seminar](#). 10/31/24.
- [WUSTL Combinatorics Seminar](#). 9/30/24.
- [AMS Spring Central Sectional Meeting. Special Session on Algebraic Methods in Graph Theory and Applications](#). 20-minute talk. 4/21/24.
- [University at Buffalo Algebra Seminar](#). 3/4/24.
- [Institute for Advanced Study in Mathematics \(IASM\) Online Geometry and Physics Seminar](#). Zhejiang University. 1/23/24.
- [International Centre for Mathematical Sciences \(ICMS\) UK–Vietnam Mathematics Joint Meeting](#). Edinburgh. 12/5/23.
- [MIT Lie Groups Seminar](#). 11/1/23.
- [AMS Fall Eastern Sectional Meeting. Special Session on Geometry, Physics and Representation Theory](#). 50-minute talk. University of Buffalo. 9/10/23.
- [Yale Geometry, Symmetry, and Physics Seminar](#). 4/24/23.
- [MIT–Harvard–MSR Combinatorics Seminar](#). 2/15/23.
- [Northeastern Geometry, Physics, and Representation Theory Seminar](#). 10/6/22.
- [ICERM Workshop on Braids in Symplectic and Algebraic Geometry](#). 10-minute talk. 3/23/22.
- [Oxford Algebra Seminar](#). 11/30/21.
- [UMass-Amherst Representation Theory Seminar](#). 10/25/21.
- [Cluster Algebra Seminar](#). [AIM Link Homology](#) research community. 10/7/21.
- [Categorical Braid Group Actions and Categorical Representation Theory](#). UMass-Amherst Focused Research Group (FRG) Conference. 6/17/21.
- [MIT Lie Groups Seminar](#). 3/3/21.

- Chinese University of Hong Kong (CUHK) Representation and Number Theory Seminar. 1/15/21.
- Harvard/MIT Algebraic Geometry Seminar. 9/29/20.
- MIT Lie Groups Seminar. 12/4/19.
- MIT Geometric Representation Theory Seminar. 12/4/19.
- Caltech Algebra and Geometry Seminar. 11/11/19.
- AMS Fall Western Sectional Meeting. [Special Session on Geometric Methods in Representation Theory](#). 20-minute talk. UC-Riverside. 11/9/19.
- UPenn Math-Physics Joint Seminar. 10/22/19.
- Hilbert Schemes, Categorification, and Combinatorics. 30-minute talk. UC-Davis FRG conference. 6/19/19.
- Perimeter Institute Mathematical Physics Seminar. 5/21/19.
- Johns Hopkins Number Theory Seminar. 4/29/19.
- UC-Davis Algebraic Geometry Seminar. 11/14/18.

Non-Research Invited Talks

- Howard University Graduate Student Seminar. 10/31/25.
- Algebraic Geometry Seminar. AIM Link Homology research community. 10/1/21.
- Geometric Categorification Seminar. AIM Link Homology research community. 7/5/21.
- MIT Independent Activities Period (IAP) Mathematics Lecture Series (18.095). 1/20/21.
- Graduate Summer School on the Geometry and Modular Representation Theory of Algebraic Groups. Simons Center for Geometry and Physics (SCGP). 30-minute talk. 8/20/19.
- Interactions between Representation Theory and Algebraic Geometry. 30-minute talk. UChicago. 8/21/17.

Conference and Workshops

- [Symmetric Functions and Knot Homology](#). New Frontiers at the Intersection of Combinatorics and Representation Theory. 3/30–4/3/27.
- [Workshop II: Contact Geometry, Cluster Algebras, and Skein Theory](#). Quantum Topology, Character Varieties and Low-Dimensional Geometry. Institute for Pure & Applied Mathematics (IPAM). 10/26–10/29/26.
- [SIAM Texas–Louisiana Sectional Meeting](#). SMU. 10/30–11/1/26.
- [AMS Fall Eastern Sectional Meeting](#). GWU. 10/3–10/4/26.
- [International Congress of Mathematicians \(ICM\) 2026](#). Philadelphia. 7/23–7/30/26.
- [Summer Workshop on Character Sheaves on Loop Groups](#). UMinnesota-Twin Cities. 6/8–6/12/26.
- [AMS Joint Mathematics Meetings \(JMM\)](#). Washington, DC. 1/4–1/7/26.
- [Webs in Algebra, Geometry, Topology, and Combinatorics](#). ICERM workshop. 12/8–12/12/25.
- [Knots in Washington 52](#). GWU conference. 12/5–12/7/25.
- [Diagrammatic Categorification](#). ICERM workshop. 10/20–10/24/25.
- [Categorification and Symplectic Duality](#). Northeastern workshop. 6/3–6/7/25.

- [Representation Theory Days](#). MIT conference in honor of George Lusztig. 11/9–11/11/24.
- [Summer School and Workshop on Relative Langlands Duality](#). UMinnesota-Twin Cities. 6/3–6/8/24.
- [AMS Spring Central Sectional Meeting](#). UW-Milwaukee. 4/20–4/21/24.
- [Monodromy and Its Applications](#). Princeton conference in honor of Nick Katz. 12/7–12/9/23.
- UK-Vietnam Mathematics Joint Meeting. ICMS conference. 12/4–12/8/23.
- [AMS Fall Eastern Sectional Meeting](#). University of Buffalo. 9/9–9/10/23.
- [Coulomb Branches and Knot Homology](#). MIT summer school. 6/19–6/23/23.
- Braids in Symplectic and Algebraic Geometry. ICERM workshop. 3/21–3/25/22.
- Link Homology. AIM online research community. 2021–22.
- Categorical Braid Group Actions. UMass FRG conference. 6/14–6/18/21.
- [AMS Fall Western Sectional Meeting](#). UC-Riverside. 11/9–11/10/19.
- [Geometry and Representation Theory of Algebraic Groups](#). SCGP summer school. 8/19–8/23/19.
- Hilbert Schemes, Categorification, and Combinatorics. UC-Davis FRG conference. 6/19–6/23/19.
- [Categorified Hecke Algebras, Link Homology, and Hilbert Schemes](#). AIM workshop. 10/1/18–10/5/18.
- [Deligne–Lusztig Theory](#). Workshop on Algebra and Representation Theory, Held on Oregonian Grounds (WARTHOG) summer school. 8/12–8/18/18.
- [Interactions between Representation Theory and Algebraic Geometry](#). UChicago conference in honor of Sasha Beilinson and Victor Ginzburg. 8/21–8/24/17.
- [Soergel Bimodules](#). MSRI (SLMath) workshop. 6/26–7/7/17.

Teaching

Course Instructor. Howard University 2025–

- MATH 199: Introduction to General Topology (2026)
- MATH 300: Graduate Seminar (2026, *2 lectures*)
- MATH 251: Topology II (2026)
- MATH 250: Topology I (2025, 2026)

Course Instructor. Yale University 2024–25

- MATH 340: Advanced Linear Algebra (2025)
- MATH 430: Introduction to Topology (2025)
- MATH 665: Topics in Quantum Algebra (2024)

Recitation Instructor. MIT 2021–24

- 18.02/18.02A: Calculus (2024, *double recitation*)
- 18.02A: Calculus (2023, *double*)
- 18.01A: Calculus (2023, *double*)
- 18.01A: Calculus (2021)
- 18.02A: Calculus (2021)

Course Instructor. MIT 2022–23

- 18.781: Theory of Numbers (2023)
- 18.708: Topics in Algebra (2022, 4 lectures)
- 18.704: Seminar in Algebra (2022)

Course Administrator. MIT 2022

- 18.01: Calculus

Graduate Instructor. University of Chicago 2017–20

- MATH 152/44: Calculus II (2020)
- MATH 153/44: Calculus III (2020)
- MATH 132/58: Elementary Functions and Calculus II (2018)
- MATH 131/58: Elementary Functions and Calculus I (2018)
- MATH 195/51: Mathematical Methods for the Social Sciences (2017)

Teaching Assistant. University of Chicago 2015–16

- MATH 242: Algebraic Number Theory (2016)
- MATH 159/53: Introduction to Proof in Linear Algebra and Analysis (2016)
- MATH 175: Elementary Number Theory (2015)

Service

Referee. 2020–

Journals have included *Advances in Mathematics*, *Geometry & Topology*, *International Mathematical Research Notices*, *Journal of the London Mathematical Society*, *Mathematische Annalen*, *Proceedings of the American Mathematical Society*, *Representation Theory*, *Science China Mathematics*, *Transformation Groups*.

Co-organizer. AMS Fall Eastern Sectional Meeting 2026

- [Special Session on Springer Fibers, Harmonic Analysis, and Arithmetic](#)

Course Coordinator. Howard University 2026–

- MATH 007: Precalculus

Graduate Admissions Committee. Howard University 2025–

Math Club Faculty Advisor. Howard University 2025–

Co-organizer. [GWU–Howard Combinatorics & Algebra Seminar](#) 2025–

Co-organizer. [Yale Geometry, Symmetry, and Physics Seminar](#) 2024–25

Research Advisor. [PROMYS](#) 2023

Mentor. [MIT SPUR](#) 2023

- [Maxim Li and Misheel Otgonbayar](#)

Mentor. [MIT PRIMES](#) 2021–23

- Michael Yang (2023)

[2024 Regeneron STS Scholar](#)

— Luke Robitaille (2021)

2022 Regeneron STS Winner, Tenth Place

2022 Davidson Fellow

Mentor. MIT UROP

2022–23

Research Advisor. MIT PRIMES

2021–23

— Ethan Liu (*Mentor:* Merrick Cai, 2023)

— Joshua Guo (*Mentor:* Kevin Chang, 2021)

2022 Regeneron STS Scholar

— Ilaria Seidel (*Mentor:* Jeffery Yu, 2021)

Mentor. UChicago Directed Reading Program

2015–20

QED Day Judge. Math Circles of Chicago

2019

Mentor. UChicago REU

2016–19

— Karhan Kayan (2019)

— Gwyn Moreland (2016)

— James Stevens (2016)

QED Day Senior Judge. Math Circles of Chicago

2016

Advising Chair. Princeton Math Club

2013

Director. Princeton University Math Competition

2011–12

Honors

— Plenary speaker at Knots in Washington 53.

— **2020 Mathematical Sciences Postdoctoral Research Fellowship (MSPRF)** from the National Science Foundation.

— **2019 William Rainey Harper Dissertation Fellowship** from the University of Chicago.

— 2014 Graduate Research Fellowship (GRFP) from the National Science Foundation.

— Phi Beta Kappa Society at Princeton University.

— Society of Sigma Xi at Princeton University.

— 2011 Shapiro Prize for Academic Excellence from Princeton University.

Profiles

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